



SQL – DDL

SHENKAR

ROAZ SCHNEIDER

CREATE TABLE

create table without data:

CREATE TABLE table (column_name column_type [(length)] [NOT NULL],...);

CREATE TABLE EXAMPLE

(

AB VARCHAR(15),

CD INTEGER NOT NULL,

EF DECIMAL(5),

GH DECIMAL(7,2)

PRIMARY KEY (AB,CD)

);

create table from query:

CREATE TABLE table AS SELECT ...;

create table actor_as_select

*select * from sakila.actor limit 10;*

```
select * from sakila.actor limit 10;
```

	actor_id	first_name	last_name	last_update
▶	1	PENELOPE	GUINNESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	GRACE	MOSTEL	2006-02-15 04:34:33
	8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
	9	JOE	SWANK	2006-02-15 04:34:33
	10	CHRISTIAN	GABLE	2006-02-15 04:34:33

```
select * from actor_as_select;
```

	actor_id	first_name	last_name	last_update
▶	1	PENELOPE	GUINNESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	GRACE	MOSTEL	2006-02-15 04:34:33
	8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
	9	JOE	SWANK	2006-02-15 04:34:33
	10	CHRISTIAN	GABLE	2006-02-15 04:34:33

The new schema of the new tables looks like:

Field	Type	Null	Key	Default
actor_id	smallint unsigned	NO		0
first_name	varchar(45)	NO		NULL
last_name	varchar(45)	NO		NULL
last_update	timestamp	NO		CURRENT_TIMESTAMP

ניתן לשנות את סדר העמודות:

```
create table actor_as_select_2
select last_name,first_name,actor_id
from sakila.actor limit 5;
```

```
create table actor_as_select_3
select last_name lname ,first_name fname ,actor_id aid
from sakila.actor limit 5;
```

CHANGE TABLE SCHEMA - ALTER

הוספת עמודה:

```
alter table actor_as_select_3 add b_day date;
```

הורדת עמודה:

```
alter table actor_as_select_3 drop b_day;
```

שינוי עמודה:

```
alter table actor_as_select_3 modify lname varchar(55);
```

DROP TABLE

מחיקת טבלה:

```
drop table table_name;
```

```
drop table actor_as_select_3;
```

insert \ update \ delete

INSERT

INSERT INTO tablename [(column-list)] VALUES (constant-list);

	actor_id	first_name	last_name	last_update
▶	1	PENELOPE	GUINESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	GRACE	MOSTEL	2006-02-15 04:34:33
	8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
	9	JOE	SWANK	2006-02-15 04:34:33
	10	CHRISTIAN	GABLE	2006-02-15 04:34:33

*insert into actor_as_select(actor_id,first_name, last_name, last_update)
values(11,'Gal','Omer','2006-02-14');*

	actor_id	first_name	last_name	last_update
▶	1	PENELOPE	GUINESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	GRACE	MOSTEL	2006-02-15 04:34:33
	8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
	9	JOE	SWANK	2006-02-15 04:34:33
	10	CHRISTIAN	GABLE	2006-02-15 04:34:33
	11	Gal	Omer	2006-02-14 00:00:00

ניתן להכניס נתונים רק לחלק מהעמודות:

```
insert into actor_as_select(actor_id,first_name,last_name)
values(12,'Stas','K');
```

	actor_id	first_name	last_name	last_update
	1	PENELOPE	GUINNESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	GRACE	MOSTEL	2006-02-15 04:34:33
	8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
	9	JOE	SWANK	2006-02-15 04:34:33
	10	CHRISTIAN	GABLE	2006-02-15 04:34:33
	11	Gal	Omer	2006-02-14 00:00:00
▶	12	Stas	K	2021-04-01 20:51:46

לעמודות שלא נתייחס בפקודת ה INSERT יכנס NULL או ערך ברירת מחדל (במידה והוגדר)
אם אחת העמודות מוגדרת כ NULL NOT ולא נכניס אליה נתון פקודת ה INSERT תכשל

ניתן להכניס נתונים לטבלה גם ב SELECT

```
insert into actor_as_select  
select * from actor_as_select limit 3;
```

	actor_id	first_name	last_name	last_update
	1	PENELOPE	GUINESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33
	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	6	BETTE	NICHOLSON	2006-02-15 04:34:33
	7	GRACE	MOSTEL	2006-02-15 04:34:33
	8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
▶	9	JOE	SWANK	2006-02-15 04:34:33
	10	CHRISTIAN	GABLE	2006-02-15 04:34:33
	11	Gal	Omer	2006-02-14 00:00:00
	12	Stas	K	2021-04-01 20:51:46
	1	PENELOPE	GUINESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33

UPDATE

UPDATE tablename SET column-assignment-list
WHERE conditional-expression ;

```
update actor_as_select  
set first_name = 'EDI', last_update = now()  
where actor_id = 3;
```

	actor_id	first_name	last_name	last_update
	1	PENELOPE	GUINESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	ED	CHASE	2006-02-15 04:34:33
	4	JENNIFER	DAVIS	2006-02-15 04:34:33



	actor_id	first_name	last_name	last_update
▶	1	PENELOPE	GUINESS	2006-02-15 04:34:33
	2	NICK	WAHLBERG	2006-02-15 04:34:33
	3	EDI	CHASE	2021-04-01 21:23:07
	4	JENNIFER	DAVIS	2006-02-15 04:34:33

ניתן לעדכן נתונים בעמודה על בסיס אותה עמודה או מעמודה אחרת

```
update actor_as_select  
set actor_id = actor_id * 2;
```

actor_id	first_name	last_name	last_update
2	PENELOPE	GUINNESS	2021-04-01 21:40:36
4	NICK	WAHLBERG	2021-04-01 21:40:36
6	EDI	CHASE	2021-04-01 21:40:36
8	JENNIFER	DAVIS	2021-04-01 21:40:36
10	JOHNNY	LOLLOBRIGIDA	2021-04-01 21:40:36
12	BETTE	NICHOLSON	2021-04-01 21:40:36

ניתן לעדכן נתונים מתוך שאילתה

```
update actor_as_select  
set last_update = (select min(last_update) from sakila.actor);
```

	actor_id	first_name	last_name	last_update
►	2	PENELOPE	GUINNESS	2006-02-15 04:34:33
	4	NICK	WAHLBERG	2006-02-15 04:34:33
	6	EDI	CHASE	2006-02-15 04:34:33
	8	JENNIFER	DAVIS	2006-02-15 04:34:33
	10	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
	12	BETTE	NICHOLSON	2006-02-15 04:34:33
	14	GRACE	MOSTEL	2006-02-15 04:34:33
	16	MATTHEW	JOHANSSON	2006-02-15 04:34:33

DELETE

DELETE FROM tablename
WHERE conditional-expression ;

*delete from actor_as_select
where actor_id = 2;*

actor_id	first_name	last_name	last_update
2	PENELOPE	GUINNESS	2006-02-15 04:34:33
2	PENELOPE	GUINNESS	2006-02-15 04:34:33
4	NICK	WAHLBERG	2006-02-15 04:34:33
4	NICK	WAHLBERG	2006-02-15 04:34:33
6	EDI	CHASE	2006-02-15 04:34:33
6	EDI	CHASE	2006-02-15 04:34:33
8	JENNIFER	DAVIS	2006-02-15 04:34:33
10	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
12	BETTE	NICHOLSON	2006-02-15 04:34:33
14	GRACE	MOSTEL	2006-02-15 04:34:33
16	MATTHEW	JOHANSSON	2006-02-15 04:34:33
18	JOE	SWANK	2006-02-15 04:34:33
20	CHRISTIAN	GABLE	2006-02-15 04:34:33
22	Gal	Omer	2006-02-15 04:34:33
24	Stas	K	2006-02-15 04:34:33



actor_id	first_name	last_name	last_update
4	NICK	WAHLBERG	2006-02-15 04:34:33
4	NICK	WAHLBERG	2006-02-15 04:34:33
6	EDI	CHASE	2006-02-15 04:34:33
6	EDI	CHASE	2006-02-15 04:34:33
8	JENNIFER	DAVIS	2006-02-15 04:34:33
10	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
12	BETTE	NICHOLSON	2006-02-15 04:34:33
14	GRACE	MOSTEL	2006-02-15 04:34:33
16	MATTHEW	JOHANSSON	2006-02-15 04:34:33
18	JOE	SWANK	2006-02-15 04:34:33
20	CHRISTIAN	GABLE	2006-02-15 04:34:33
22	Gal	Omer	2006-02-15 04:34:33
24	Stas	K	2006-02-15 04:34:33

`delete from actor_as_select
where actor_id not in (select actor_id from actor_as_select_2);`

actor_id	first_name	last_name	last_update
4	NICK	WAHLBERG	2006-02-15 04:34:33
4	NICK	WAHLBERG	2006-02-15 04:34:33
6	EDI	CHASE	2006-02-15 04:34:33
6	EDI	CHASE	2006-02-15 04:34:33
8	JENNIFER	DAVIS	2006-02-15 04:34:33
10	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
12	BETTE	NICHOLSON	2006-02-15 04:34:33
14	GRACE	MOSTEL	2006-02-15 04:34:33
16	MATTHEW	JOHANSSON	2006-02-15 04:34:33
18	JOE	SWANK	2006-02-15 04:34:33
20	CHRISTIAN	GABLE	2006-02-15 04:34:33
22	Gal	Omer	2006-02-15 04:34:33
24	Stas	K	2006-02-15 04:34:33



actor_id	first_name	last_name	last_update
4	NICK	WAHLBERG	2006-02-15 04:34:33
4	NICK	WAHLBERG	2006-02-15 04:34:33

EXERCISES

1. In sand_box database create new table: city based on city from database sakila. The new table contains all cities from Spain (city, country).
2. Create also country table and insert into it the data for Spain.
3. On the new tables update the country_id for Spain to 1.
4. Add column to city table – city_code (varchar 10) and update values for it – city_id + “_” + country_id (use concat function).
5. Delete from the new city table the city with the name - Santiago de Compostela
6. Create a new table city_backup as a copy of the new city table structure only (no data).
Then insert into it only cities where the name starts with ‘B’ from sakila.city.
7. Rename city_backup to city_b.
8. Drop all new tables.

